DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW

EVALUATION SCHEME & SYLLABUS

FOR

B. TECH. THIRD YEAR

- ELECTRONICS AND INSTRUMENTATION ENGINEERING
- INSTRUMENTATION AND CONTROL ENGINEERING
- APPLIED ELECTRONICS AND INSTRUMENTATION
- INSTRUMENTATION ENGINEERING

AS PER

AICTE MODEL CURRICULUM

[Effective from the Session: 2024-25]

B.Tech. V Semester (Instrumentation and Control Engineering)

S. No.	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total SW+ESE	Credit
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BEC501	Integrated Circuits	T	PC	3	1	0	20	10	30	70	100	4
2	BEC502	Microprocessor & Microcontroller	T	PC	3	1	0	20	10	30	70	100	4
3	BIC-501	Instrumentation & Measurement	T	PC	3	1	0	20	10	30	70	100	4
4	BIC051/ BEC052/ BEC053/ BEC054	Department Elective-I	T	PC	3	0	0	20	10	30	70	100	3
5	BIC052/ BIC053/ BEC057/ BIC054	Department Elective-II	T	PC	3	0	0	20	10	30	70	100	3
6	BEC551	Integrated Circuits Lab	P	PC	0	0	2		50	50	50	100	1
7	BEC552	Microprocessor & Microcontroller Lab	P	PC	0	0	2		50	50	50	100	1
8	BIC-551	Instrumentation & Measurement Lab	P	PC	0	0	2		50	50	50	100	1
9	BEC-554	Mini Project/Internship **	P	PC	0	0	2		50	50		50	1
10	BNC501/ BNC502	Constitution of India/ Essence of Indian Traditional Knowledge	T	VA	2	0	0	20	10	30	70	100	NC
11		Honors Degree / Minor Degree Courses											
		Total										**850**	**22**

****The Mini Project or Internship (4weeks) conducted during summer break after IV Semester and will be assessed during Vth Semester.**

Course Code	Course Title
	Department Elective-I
BIC-051	Transducer and Sensor Measurement System
BEC-052	Bio-Medical Sensors & Instrumentation
BEC-053	Intelligent Systems and Robotics
BEC-054	VLSI Technology
	Department Elective-II
BIC-052	Digital Signal Processing
BIC-053	Drone Technology
BEC-057	Optical Communication
BIC-054	Bio-Instrumentation

B.Tech. VI Semester
Instrumentation and Control Engineering

S. No.	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total SW+ESE	Credit
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BIC601	Embedded System Design	T	PC	3	1	0	20	10	30	70	100	4
2	BEC602	Control System	T	PC	3	1	0	20	10	30	70	100	4
3	BIC602	Industrial Measuring Instruments	T	PC	3	1	0	20	10	30	70	100	4
4	BIC061/ BEC062/ BIC062/ BIC063	Department Elective–III	T	PC	3	0	0	20	10	30	70	100	3
5	----	Open Elective-I	T	PC	3	0	0	20	10	30	70	100	3
6	BIC651	Embedded System Design Lab	P	PC	0	0	2		50	50	50	100	1
7	BEC652	Control System Lab	P	PC	0	0	2		50	50	50	100	1
8	BIC652	Industrial Measuring Instruments Lab	P	PC	0	0	2		50	50	50	100	1
9	BNC601/ BNC602	Constitution of India/ Essence of Indian Traditional Knowledge	T	VA	2	0	0	20	10	30	70	100	NC
10		Honors Degree / Minor Degree Courses											
		Total										**800**	**21**

Course Code	Course Title
	Department Elective-III
BIC-061	Digital Measurement Techniques
BEC-062	Data Communication Networks
BIC-062	Bio-Medical Signal Processing
BIC-063	Microcontrollers and Sensors for IoT Devices

B.Tech 3rd Year
V Semester
Syllabus

BEC -501	INTEGRATED CIRCUITS	3L:1T:0P	4 Credits

Unit	Topics	Lectures
I	**The 741 IC Op-Amp**: General operational amplifier stages (bias circuit, the input stage, the second stage, the output stage, short circuit protection circuitry), device parameters, DC and AC analysis of input stage, second stage and output stage, gain, frequency response of 741, a simplified model, slew rate, relationship between ft and slew rate.	8
II	**Linear Applications of IC Op-Amps:** Op-Amp based V-I and I-V converters, instrumentation amplifier, generalized impedance converter, simulation of inductors. **Active Analog filters:** Sallen Key second order filter, Designing of second order lowpass and high pass Butterworth filter, Introduction to band pass and band stop filter, all pass active filters, KHN Filters. Introduction to design of higher order filters.	8
III	**Frequency Compensation & Nonlinearity:** Frequency Compensation, Compensation of two stage Op-Amps, Slewing in two stage Op-Amp. Nonlinearity of Differential Circuits, Effect of Negative feedback on Nonlinearity. **Non-Linear Applications of IC Op-Amps:** Basic Log–Anti Log amplifiers using diode and BJT, temperature compensated Log-Anti Log amplifiers using diode, peak detectors, sample and hold circuits. Op-amp as a comparator and zero crossing detector, astable multivibrator & monostable multivibrator. Generation of triangular waveforms, analog multipliers and their applications.	4 8
IV	**Digital Integrated Circuit Design:** An overview, CMOS logic gate circuit's basic structure, CMOS realization of inverters, AND, OR, NAND and NOR gates. **Latches and Flip flops**: the latch, CMOS implementation of SR flip-flops, a simpler CMOS implementation of the clocked SR flip-flop, CMOS implementation of J-K flip- flops, D flip-flop circuits.	6
V	**Integrated Circuit Timer:** Timer IC 555 pin and functional block diagram,Monostable and Astable multivibrator using the 555 IC. **Voltage Controlled Oscillator**: VCO IC 566 pin and functional block diagram and applications. **Phase Locked Loop (PLL)**: Basic principle of PLL, block diagram, working, Ex-ORgates and multipliers as phase detectors, applications of PLL.	6

Text Book:

1. Microelectronic Circuits, Sedra and Smith, 7th Edition, Oxford, 2017.
2. Behzad Razavi: Design of Analog CMOS Integrated Circuits, TMH

Reference Books:

1. Gayakwad: Op-Amps and Linear Integrated Circuits, 4th Edition Prentice Hall of India, 2002.
2. Franco, Analog Circuit Design: Discrete & Integrated, TMH, 1st Edition.
3. Salivahnan, Electronics Devices and Circuits, TMH, 3rd Edition, 2015
4. Millman and Halkias: Integrated Electronics, TMH, 2nd Edition, 2010

Course Outcomes: At the end of this course, Students will be able to:

1. Explain complete internal analysis of Op-Amp 741-IC.
2. Examine and design Op-Amp based circuits and basic components of ICs such as various types of filters.
3. Implement the concept of Op-Amp to design Op-Amp based non-linear applications and wave-shaping circuits.
4. Analyse and design basic digital IC circuits using CMOS technology.
5. Describe the functioning of application specific ICs such as 555 timer, VCO IC 566 and PLL.

BEC-502	MICROPROCESSOR & MICROCONTROLLER	3L:1T:0P	4 Credits

Unit	Topics	Lectures
I	**Introduction to Microprocessor:** Microprocessor architecture and its operations, Memory, Input & output devices, The 8085 MPU- architecture, Pins and signals, Timing Diagrams, Logic devices for interfacing, Memory interfacing, Interfacing output displays, Interfacing input devices, Memory mapped I/O.	8
II	**Basic Programming concepts**: Flow chart symbols, Data Transfer operations, Arithmetic operations, Logic Operations, Branch operation, writing assembly language programs, Programming techniques: looping, counting and indexing. Additional data transfer and 16-bit arithmetic instruction, Logic operation: rotate, compare, counter and time delays, 8085 Interrupts.	8
III	**16-bit Microprocessors (8086):** Architecture, Pin Description, Physical address, segmentation, memory organization, Addressing modes. **Peripheral Devices:** 8237 DMA Controller, 8255 programmable peripheralinterface, 8253/8254programmable timer/counter, 8259 programmable interrupt controller, 8251 USART and RS232C.	8
IV	**8051 Microcontroller Basics**: Inside the Computer, Microcontrollers and Embedded Processors, Block Diagram of 8051, PSW and Flag Bits, 8051 Register Banks and Stack, Internal Memory Organization of 8051, IO Port Usage in 8051, Types of Special Function Registers and their uses in 8051, Pins Of 8051. Memory Address Decoding, 8031/51 Interfacing with External ROM And RAM. 8051 Addressing Modes.	8
V	**Assembly programming and instruction of 8051:** Introduction to 8051 assembly programming, Assembling and running an 8051 program, Data types and Assembler directives, Arithmetic, logic instructions and programs, Jump, loop and call instructions, IO port programming. Programming 8051 Timers. Serial Port Programming, Interrupts Programming, **Interfacing:** LCD & Keyboard Interfacing, ADC, DAC & Sensor Interfacing, External Memory Interface, Stepper Motor and Waveform generation.	8

Text Books:

1. Ramesh Gaonkar, "Microprocessor Architecture, Programming, and Applications with the 8085", 6th Edition, Penram International Publication (India) Pvt. Ltd.,2013
2. D. V. Hall: Microprocessors Interfacing, TMH 3rd Edition,
3. Mazidi Ali Muhammad, Mazidi Gillispie Janice, and McKinlay Rolin D., "The 8051 Microcontroller and Embedded Systems using Assembly and C", Pearson, 2nd Edition,2006

Reference Books:

1. Kenneth L. Short, "Microprocessors and programmed Logic", 2nd Ed, Pearson Education Inc.,2003
2. Barry B. Brey, "The Intel Microprocessors, 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, Pentium-Pro Processor, Pentium-II, Pentium-III, Pentium-IV, Architecture, Programming & Interfacing", Eighth Edition, Pearson Prentice Hall, 2009.
3. Shah Satish, "8051 Microcontrollers MCS 51 Family and its variants", Oxford,2010

Course Outcomes: At the end of this course, Students will be able to:

1. Demonstrate the basic architecture of 8085.
2. Illustrate the programming model of microprocessors & write program using 8085microprocessors.
3. Demonstrate the basics of 8086 Microprocessor and interface different external Peripheral Devices like timer, USART etc. with Microprocessor (8085/8086).
4. Compare Microprocessors & Microcontrollers, and comprehend the architecture of 8051 microcontroller
5. Illustrate the programming model of 8051 and implement them to design projects on real time problems.

BIC-501	MEASUREMENTS & INSTRUMENTATION	3L:1T:0P	4 Credits

Unit	Topics	Lectures
I	**Electrical Measurements:** Measurement system, Characteristics of instruments, Methods of measurement, Errors in Measurement & Measurement standards, Measurement error combination, **Review of indicating and integrating instruments:** PMMC instrument, Galvanometer, DC ammeter, DC voltmeter, Series ohm meter.	8
II	**Electronic Instruments:** Transistor voltmeter circuits, AC electronic voltmeter, current measurement with electronic instruments, probes, **Digital voltmeter systems**: Digital multimeter, digital frequency meter System, Oscilloscope **Instrument calibration:** Comparison method, digital multimeter as standard instrument, Calibration instrument.	8
III	**Measuring Methods:** Voltmeter and Ammeter methods, Wheatstone bridge, Measurement of low, medium and high resistances, Insulation resistance measurement, AC bridges for inductance and capacitance measurement, Q meter.	8
IV	**Electronic Measurements:** Electronic instruments: Wattmeter & Energymeter. Time, Frequency and phase angle measurements using CRO; Storage oscilloscope, Spectrum & Wave analyzer, Digital counter, Frequency meter.	8
V	**Instrumentation**: Transducers, classification & selection of transducers, strain gauges, Thermistors, Thermocouples, LVDT, Inductive & capacitivetransducers, Piezoelectric and Hall-effect transducers, Measurement of motion, force, pressure, temperature, flow and liquid level. Concept of signal conditioning and data acquisition systems, Concept of smart sensors and virtual instrumentation.	8

Text Books:

1. A K Sawhney, "Electrical & Electronic Measurement & Instrument", Dhanpat Rai & Sons, India (2015).
2. BC Nakra & K. Chaudhary, "Instrumentation, Measurement and Analysis," TMH, 2nd Edition (2009).

Reference Books:

1. WD Cooper, "Electronic Instrument & Measurement Technique", Prentice HallInternational (2001).
2. E. O. Doebelin, "Measurements systems: Applications and Design", 6th Edition,Tata McGraw Hil 2017.

Course Outcomes: At the end of this course, Students will be able to:

1. Classify the Instrumentation and Measurement system and various measurementerrors.
2. Analyze and design voltmeter circuits, AC electronic voltmeter, digital frequencymeter and current measurement with electronic instruments.
3. Evaluate various resistance and impedance measuring methods using Bridges andQ-meter.
4. Analyze fundamental operation of CRO and some special type of oscilloscopes likeDSO, Sampling oscilloscope.
5. Demonstrate calibration method to calibrate various instruments and classifytransducers like for force, pressure, motion, temperature measurement etc.

BIC-051	TRANSDUCER AND SENSOR MEASUREMENT SYSTEM	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**GENERALISED CONFIGURATIONS, FUNCTIONAL DISCRIPTION & PERFORMANCE CHARACTERISTICS OF MEASURING INSTRUMENTS:** Functional elements of an instrument; active & passive transducers; analog & digital modes of operation; null & deflection methods; I/O configuration of measuring instruments & instrument system – methods of correction for interfering & modifying inputs. Static characteristics; Meaning of static calibration, accuracy, precision & bias. Combination of component errors in overall system-accuracy calculation. Static sensitivity, linearity, threshold, resolution, hysteresis and dead space. Scale readability. Span. Generalized static stiffness & input impedance. Computer aided calibration & measurement, multiple regressions.	8
II	**MEASUREMENT OF DISPLACEMENT, FORCE, TORQUE & SHAFT POWER:** Principle of measurement of displacement. Resistive potentiometers, variable inductance & variable reluctance pickups, LVDT, capacitance pickup. Principle of measurement of Force, Torque, Shaft power standards & calibration; basicmethods of force measurement; characteristics of elastic force transducer- Bonded strain gauge, differential transformer, piezo electric transducer, variable reluctance/FM-oscillator, digital systems. Loading effects; Torque measurement on rotating shafts, shaft power measurement (dynamometers).	8
III	**TEMPERATURE MEASUREMENT**: Standards & calibration; thermal expansion methods- bimetallic thermometers, liquid-in-glass thermometers, pressure thermometers; thermoelectric sensor (thermocouple) – common thermocouple, special materials, configuration & techniques; electrical resistance sensors – conductive sensor (resistance thermometers), bulk semiconductor sensors (thermistors), bulk semiconductor sensors (thermistors); junction semiconductor sensors; digital thermometers. Radiation Methods – radiation fundamentals, radiation detectors, unchopped (dc) broadband radiation thermometers. Chopped (AC) selective band(photon) radiation thermometers, automatic null balance radiation thermometers (optical pyrometers). Two color radiation thermometers, Black body-tipped fibre optic radiation thermometer, IR imaging systems. Fluoroptic temperature measurement.	8
IV	**PRESSURE MEASUREMENT:** Standards & calibration; basic methods of pressure measurement; dead weight gauges & manometer, manometer dynamics; elastic transducers; high pressure measurement; low pressure (vaccum) measurement – Mcleod gage, Knudsen gage, momentum-transfer (viscosity) gages, thermal conductivity gages, ionization gages, dual gage technique.	8
V	**FLOW MEASUREMENT:** Local flow velocity, magnitude and direction. Flow visualization. Hot wire and hot film anemometer. Hot-film shock-tube velocity sensor. Laser Doppler velocity meter; gross volume flow rate: calibration and standards. Constant-area, variable-pressure-drop meters (obstruction meters). Averaging pilot tubes. Constant pressure drop, variable area meters (rotameters), turbine meters, and positive displacement meters. Metering pumps. Electromagnetic flow meters. Drag force flow meters. Ultrasonic flow meters, vortex shedding flow meters. **LEVEL MEASUREMENT:** Capacitance probe; conductivity probes; diaphragm level detector, deferential pressure level detector, radiation level sensors, RADAR level gauges, level transmitter, ultrasonic level detector.	8

Text Books:

1. Transducers and Instrumentation – D. V. S. Murty, 2nd Edition, PHI, 2009
2. Instrumentation Measurement and Analysis- B. C. Nakra and K. K. Choudhry,3rd Edition, McGraw Hill Education (India) Pvt. Ltd. 2009

Reference Books:

1. Measurement systems application and design, ERNEST DOEBELIN, IV Edn.
2. Introduction to Measurements and Instrumentation – A. K. Ghosh, 2nd Edition,PHI, 2007.

Course Outcomes: At the end of this course, Students will be able to:

1. Explain the working of measurement systems and different types of sensors andtransducers.
2. Formulate the sensor to measure various physical parameters used in Industry and normal measurement applications.
3. Analyze the working principle of resistive, inductive and capacitive transducers and their applications.
4. Differentiate the thermocouples, piezoelectric and pyro-electric transducers and apply them in various applications.
5. Describe acoustic, optical sensors and other sensors and their applications.

BEC-052	BIO-MEDICAL SENSORS & INSTRUMENTATION	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Generalized configurations, functional descriptions, and performance characteristics of measuring instruments:** General concepts and terminology of measurement systems, transducer classification, general input-output configuration, static and dynamiccharacteristics of a measurement system, Statistical analysis of measurement data. Standards and Calibration.	8
II	**Introduction of Bio-medical Instrumentation, Sources of Bioelectric Potentials and Electrodes**: Introduction to man-instrument system, components of the man-instrument system, Physiological system of the body, Problems encountered in measuring a living system. Resting and action potentials, Propagation of action potentials, bioelectric potentials, Bio potential electrodes, Biochemical transducers. Review of transducers.	8
III	**Cardiovascular System and Measurements:** The heart and cardiovascular system, ECG, blood pressure and its measurement, respiration and pulse rate, characteristics and measurement of blood flow meter, cardiac output, plethysmography, pacemaker, defibrillators, heart sounds and its measurement.	8
IV	**Respiratory and Neuro-muscular System:** The physiology of the respiratory system, test and instrument for the mechanics of breathing, the somatic nervous system, EEG, EMG and GSR.	8
V	**Measurement and Recording of Non-invasive Diagnostic Instrumentation, Patient Care and Electrical Safety:** Principle of ultrasonic measurement, ultrasonic, thermography, elements of intensive care monitoring, X-ray, CT – Scan and MRI, tonometer, dialysis**,** diathermy, Shock hazards from electrical equipment.	8

Textbooks

1. Cromwell, L. and Weibell, F.J. and Pfeiffer, E.A., Biomedical Instrumentation and Measurement, Dorling Kingsley (2006) 2nd ed.
2. Carr, J.J. and Brown, J.M., Introduction to Biomedical Equipment Technology, Prentice Hall (2000) 4th ed.

Reference Books

1. Geddes, L.A., and Baker, L.E., Principles of Applied Biomedical Instrumentation, Wiley Inter-Science (1989) 3rd ed.
2. Khandpur, R.S., Handbook of Biomedical Instrumentation, McGraw Hill (2003) 2nd ed.
3. Webster, J.G., Medical Instrumentation Application and Design, John Wiley (2007) 3rd ed.

Course Outcomes: At the end of this course, Students will be able to:

1. Understand the basics of measurement system.
2. Differentiate and analyse the biomedical signal sources
3. Elucidate cardiovascular system and related measurements
4. Exhibit the knowledge of working principle and applications of the respiratory and nervous related measurements.
5. Measure the parameters non-invasive diagnostic.

BEC -053	INTELLIGENT SYSTEMS AND ROBOTICS	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**System Modeling:** Biological and Cognitive Paradigms for Robot Design, Declarative-Procedural-Reflexive Hierarchy for Decision-Making and Control, Articulated Robots, Joint-Link (Denavit - Hartenberg) Transformations, Mobile Ground Robots, Uninhabited Air Vehicles, Intelligent Agents.	8
II	**Control System Principles:** Open- and Closed-Loop Control, Time-domain and Frequency-domain Analysis, Optimality and Constraints, Stability and Performance, Adaptation, Control Actuation, Closed-form and Probabilistic Path Planning.	8
III	**Computing, Measurement, State, and Parameter Estimation:** Sensors and Sensing, Formal and Fuzzy Logic, Turing Machines and Concepts of Machine Learning, Analog and Digital Systems, Probability and Error Models, Sensor-Based Estimation, Extended Kalman and Particle Filters, Simultaneous Location and Mapping (SAM).	8
IV	**Decision-Making and Machine Learning:** Decision Trees, Bayesian Belief Networks, Classification of Data Sets, Task Planning for Individual and Multiple Agents.	8
V	**Numerical Methods for Evaluation and Search:** Monte Carlo Simulation, Genetic Algorithms, Simulated Annealing, Particle Swarm Optimization, Production Systems, Forward Chaining, Backward Chaining, Training and Implementation of Network Architectures, Feed-Forward Networks, Associative Networks, Cerebellar Model Articulation Controller, Deep-Learning Algorithms.	8

Text Books:

1. H. Asada and J.-J. Slotine, Robot Analysis and Control, J. Wiley & Sons, 1986.C. Asfahl, Robots and Manufacturing Automation, J. Wiley & Sons, 1992.
2. D. Auslander, J. Ridgely, and J. Ringgenberg, Control Software for Mechanical Systems, Prentice-Hall, 2002.
3. G. Bekey, Autonomous Robots, MIT Press, 2005.
4. Albus, J. I., and Meystel, A. M., Engineering of Mind, J. Wiley & Sons, 2001.
5. P. Antsaklis and K. Passino, An Introduction to Intelligent and Autonomous Control, Kluwer, 1993.

Reference Books:

1. J. Craig, Introduction to Robotics Mechanics and Control, Pearson, 2018
2. R. Dorf, Robotics and Automated Manufacturing, Reston (Prentice-Hall), 1983
3. D. Goldberg, Genetic Algorithms in Search, Optimization, and Machine Learning, Addison-Wesley, 1989.

Course Outcomes: At the end of this course, Students will be able to:

1. Apply the concept of Open- and Closed-Loop Control.
2. Apply Fourier to understand Time-domain and Frequency-domain Analysis
3. Apply the fundamental concepts of Closed-form and Probabilistic Path Planning
4. Analyze Formal and Fuzzy Logic.
5. Analyze Simultaneous Location and Mapping (SAM).

BEC 054	VLSI TECHNOLOGY	3L:0T:0P	3 Credits

Unit	Topic	Lectures
I	Introduction To IC Technology: SSI, MSI, LSI, VLSI Integrated Circuits. Crystal Growth and Wafer Preparation: Electronic Grade Silicon, Czochralski Crystal Growth, Silicon Shaping, Processing Considerations. Wafer Cleaning Technology - Basic Concepts, Wet cleaning, Dry cleaning	8
II	**Epitaxy:** Vapor-Phase Epitaxy, Molecular Beam Epitaxy, Silicon on Insulators, Epitaxial Evaluation. **Oxidation:** Growth Kinetics, Thin Oxides, Oxidation Techniques and Systems, Oxides Properties.	8
III	**Lithography:** Optical Lithography, Electron beam lithography, Photomasks, Wet Chemical Etching. **Dielectric and Polysilicon Film Deposition:** Deposition Processes of Polysilicon, Silicon Dioxide, Silicon Nitride.	8
IV	**Diffusion:** Models of diffusion in solids, Fick's 1-Dimensional diffusion equation, Diffusion of Impurities in Silicon and Silicon Dioxide, Diffusion Equations, Diffusion Profiles, Diffusion Furnace, Solid, Liquid and Gaseous Sources, **Ion-Implantation:** Ion-Implantation Technique, Range Theory, Implantation Equipment.	8
V	**Metallization:** Metallization Application, Metallization Choices, Physical Vapor Deposition, Vacuum Deposition, Sputtering Apparatus. **Packaging of VLSI devices**: Package Types, Packaging Design Consideration, VLSI Assembly Technologies, Package Fabrication Technologies, CMOS fabrication steps.	8

Content beyond the Course:
Topics such as IC fabrication process steps including lithography, thin film deposition (silicon dioxide, silicon nitride, polysilicon), diffusion, ion-implantation, metallization, and CMOS process integration, as covered in this syllabus, can be implemented and verified using "Siemens EDA (Calibre)/ Cadence/ Synopsys or similar relevant software tools".

Text Books:
1. S. M. Sze, "VLSI Technology", McGraw Hill Publication, 2nd Edition 2017
2. S.K. Ghandhi, "VLSI Fabrication Principles", Willy-India Pvt. Ltd, 2008

Reference Books:
1. J. D. Plummer, M. D. Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modeling", Pearson Education Publication, 2009
2. Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013

COURSE OUTCOME: After completion of the course, student will be able to:
1. Interpret the basics of crystal growth, wafer preparation and wafer cleaning.
2. Evaluate the process of Epitaxy and oxidation.
3. Differentiate the lithography, etching and deposition process.
4. Analyze the process of diffusion and ion implantation
5. Express the basic process involved in metallization and packaging.

BIC -052	DIGITAL SIGNAL PROCESSING	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**LTI Systems:** Definition, representation, impulse response, derivation for the output sequence, concept of convolution, graphical, analytical and overlap add methods to compute convolution supported with examples, properties of convolution, interconnections ofLTI systems with physical interpretations, stability and causality conditions, recursive and non-recursive systems.	8
II	**Realization of Digital Linear Systems:** IIR Filter Realization: Direct form, cascade realization, parallel form realization, Ladder structures- continued fraction expansion of H(z). FIR Filter Realization: Direct, Cascade, FIR Linear Phase Realization	8
III	**Discrete Fourier Transform:** Concept and relations for DFT/IDFT, Twiddle factors and their properties,computational burden on direct DFT, DFT/IDFT as linear transformations,DFT/IDFT matrices, computation of DFT/IDFT by matrix method, multiplication of DFTs, circular convolution, computation of circular convolution by graphical,DFT/IDFT and matrix methods, linear filtering using DFT, aliasing error, filteringof long data sequences – Overlap-Save and Overlap-Add methods with examples. **Fast Fourier Transform:** Radix-2 algorithm, decimation-in-time, decimation-in-frequency algorithms, signalflow graphs, Butterflies, computations in one place, bit reversal, examples for DIT & DIF FFT Butterfly computations with examples.	8
IV	**Filter Design:** Basic concepts of IIR and FIR filters, difference equations, design of Butterworth IIR analog filter using impulse invariant and bilinear transforms, frequency transformation. Design of FIR filters using windowing technique, Examples of FilterDesigns Using Windows (Rectangular and Hamming windows).	8
V	**Digital Signal Processors:** Introduction, Architecture, Features, Addressing Formats, Functional modes. Introduction to Commercial Digital Signal Processors.	8

Text Books:

1. John G Prokias, Dimitris G Manolakis, Digital Signal Processing. Pearson , 4th Edition, 2007
2. Johnny R. Johnson, Digital Signal Processing, PHI Learning Pvt Ltd., 2009.
3. S. Salivahanan, A. Vallavaraj, Digital Signal Processing, TMH, 4th Edition 2017.

Reference Books:

1. Oppenheim & Schafer, Digital Signal Processing. Pearson Education 2015
2. S.K. Mitra, 'Digital Signal Processing–A Computer Based Approach, TMH, 4th Edition.

Course Outcomes: At the end of this course, Students will be able to:

1. Design and describe different types of realizations of digital systems (IIR and FIR) and their utilities.
2. Select design parameters of analog IIR digital filters (Butterworth and Chebyshev filters) and implement various methods such as impulse invariant transformation and bilinear transformation of conversion of analog to digital filters.
3. Design FIR filter using various types of window functions.
4. Define the principle of discrete Fourier transform & its various properties and concept of circular and linear convolution. Also, students will be able to define and implement FFT i.e.a fast computation method of DFT.
5. Define the concept of decimation and interpolation. Also, they will be able to implement itin various practical applications.

BIC-053	DRONE TECHNOLOGY	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Introduction to Drones:** History of UAV drones, Introduction to the structure of unmanned aerial vehicles, understand different drone parts and their contribution for successful flight operation, Understand different types of drone circuits /electronic parts, classification of drones, System Composition, applications.	8
II	**Design of UAV Drone Systems:** Introduction to Design and Selection of the System, Aerodynamics and Airframe Configurations, Characteristics of Aircraft Types, Design Standards and Regulatory Aspects-India Specific, Design for Stealth.	8
III	**Avionics Hardware of Drones:** Autopilot, AGL-pressure sensors servos-accelerometer –gyros-actuators- power supply-processor, integration, installation, configuration.	8
IV	**Communication, Payloads and Controls:** Payloads, Telemetry, Tracking, controls-PID feedback, radio control frequency range, modems, memory system, simulation, ground test-analysis-trouble shooting.	8
V	**Navigation and Testing:** Waypoints navigation, ground control software, System Ground Testing, System In-flight Testing, Future Prospects and Challenges	8

Text Books:

1. Reg Austin "Unmanned Aircraft Systems UAV design, development and deployment", Wiley, 2010.
2. Robert C. Nelson, Flight Stability and Automatic Control, McGraw-Hill, Inc, 1998.
3. Kimon P. Valavanis, "Advances in Unmanned Aerial Vehicles: State of the Art and the Road to Autonomy", Springer, 2007
4. Paul G Fahlstrom, Thomas J Gleason, "Introduction to UAV Systems", UAV Systems, Inc, 1998
5. Dr. Armand J. Chaput, "Design of Unmanned Air Vehicle Systems", Lockheed Martin Aeronautics.

Reference Books / Other Resources

1. Syed Omar Faruk Towaha, Building Smart Drones with ESP8266 and Arduino: Build exciting drones by leveraging the capabilities of Arduino and ESP8266, Packt Publishing, 2018.
2. www.asteria.co.in
3. www.kisspng.com
4. www.droneaviation.com
5. www.dji.com
6. www.aerosociety.com

Course Outcomes: At the end of this course, Students will be able to:

1. Ability to design UAV drone system
2. To understand working of different types of engines and its area of applications.
3. To understand static and dynamic stability dynamic instability and control concepts
4. To know the loads taken by aircraft and type of construction and also construction materials in them.

BEC -057	OPTICAL COMMUNICATION	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Introduction to Optical Communication:** Optical Spectral Band with Operating Windows, General Communication System, Optical Communication System with its advantages. **Optical Fiber Waveguides:** Ray Theory of Transmission with TIR, Acceptance Angle, Numerical Aperture and Skew Rays, Electromagnetic Mode Theory for OpticalPropagation, Modes in a Planar Guide, Phase and Group Velocity, Phase Shift with Total Internal Reflection, Evanescent Field, Goos-Haenchen Shift, Cylindrical Fiber Modes, Mode Coupling, Step Index fibers Vs Graded Index fibers, Single Mode Fibers- Cut off wavelength, MFD & Spot Size.	8
II	**Signal Loss in Optical Fibers:** Attenuation, Material Absorption Losses (Intrinsic and Extrinsic absorption), types of Linear and Non-Linear Scattering Losses, Fiber Bending Losses, Kerr Effect. **Dispersion:** Introduction with its types: Chromatic / Intramodal Dispersion (Material and Waveguide Dispersion), Intermodal dispersion (for MSI and MGI fibers), Overall (Total) Fiber Dispersion in Multimode and Singe Mode Fiber, Dispersion Modified Single Mode Fibers, Polarization & Fiber Birefringence.	8
III	**Optical Sources: LEDs-**Introduction to LEDs & Materials used for fabrication, LED Power and Efficiency, LED Structures, LED Characteristics, Modulation Bandwidth. **Laser Diodes-**Introduction, Optical Feedback & Laser Oscillations, Resonant Frequencies, Laser Modes, and Threshold Condition for Laser Oscillation, Laser Diode Rate Equations, Semiconductor injection Laser- Efficiency, Laser Single Mode operation, Reliability of LED & ILD.	8
IV	**Power Launching in Fiber:** Source to Fiber Power Launching and Coupling Techniques, Power Launching Vs Wavelength, Equilibrium Numerical Aperture. **Photo Detectors:** Introduction, Physical Principles of Photodiodes: The PIN Photo Detector, Avalanche Photodiodes, Temperature Effect on Avalanche Gain, Detector Response Time, Photo Detector Noise: Noise Sources, Signal to Noise Ratio, Comparison of Photo Detectors, Fundamental Receiver Operation with Digital Signal Transmission.	8
V	**Digital Receiver Performance:** Probability of Error / BER, Receiver Sensitivity & The Quantum Limit, Error Control Techniques, Eye Diagram Pattern Features, Coherent Detection: Homodyne Detection and Heterodyne Detection, Digital links: Point to Point Links, Power Penalties, Multichannel & Multiplexing Transmission Techniques, basic concept of Free Space Optics (FSO) based Communication System.	8

Text Book:

1. John M. Senior, "Optical Fiber Communications", Pearson, 3rd Edition, 2010.
2. Gerd Keiser, "Optical Fiber Communications", McGraw Hill, 5th Edition, 2013.
3. Govind P. Agrawal, "Fiber Optic Communication Systems", John Wiley, 3rd Edition, 2004.

Course Outcomes: At the end of this course, Students will be able to:

1. Define and explain the basic concepts and theory of optical communication.
2. Describe the signal losses with their computation and dispersion mechanism occurring inside the optical fiber cable.
3. Differentiate the optical sources used in optical communication with their comparative study.
4. Identify different optical components on receiver side; assemble them to solve real world problems related to optical communication systems.
5. Evaluate the performance of an optical receiver to get idea about power budget and ultimately be an engineer with adequate knowledge in optical domain.

BIC-054	BIO-INSTRUMENTATION	3L:0T:0P	3 CREDITS

Unit	Topics	Lectures
I	Transducers: Classification, selection of transducers, circuit based on transduction. Temperature transducers – Displacement transducer - Pressure transducer - catheter tip transducers. Photoelectric transducers - Flow transducers - Piezoelectric transducers and their applications, Biological receptors and receptor characteristics.	8
II	Reference electrodes: The PO2 electrodes, Membrane electrodes, Blood gas analysis, Transcutaneous PO2 and PCO2 transducers, Fiber optic chemical transducer, Ion specific electrodes, Ionic content of blood, ISFET for glucose, urea.	8
III	Electrocardiogram (ECG), Electroencephalogram (EEG), Electromyogram (EMG), Electrooculogram (EOG), Electroretinogram (ERG), Recording Electrodes – Electrod tissue interface, polarization, skin contact impedance, motion artifacts, Silver-Silve Chloride electrodes, Electrodes for ECG, Electrodes for EEG, Electrodes of EMG Electrical conductivity of electrode jellies and creams, microelectrodes, Needle electrodes.	8
IV	Principle of dialysis, Artificial kidney, Types, function and working of dialysis, Performance analysis of dialysis, Membranes used for haemodialysis, Block diagram and working of haemodialysis machine, Portable kidney machine – working and flow diagram.	8
V	Physiological effects of electrical currents, macroshock and microshock, preventive measures to reduce shock hazards, Leakage current, isolation of patient circuits, safety of electrically susceptible patients, radiation hazards and safety, shielding, open ground problem and earthing methods.	8

Text Book:

1. R. S. Khandpur, Biomedical Instrumentation Technology and Applications, McGraw-Hill Professional, 2004
2. Leslie Cromwell, Fred. J. Weibell and Erich. A. Pfeiffer, "Biomedical Instrumentation and Measurements", 2nd Edition, PHI, 2003. (UNIT I, III)
3. John G. Webster, Medical Instrumentation: Application and Design, 3rd edition, John Wiley & Sons, New York, 1998. (UNIT II)
4. Raja Rao, C, Guha, S.K, Principles of Medical Electronics and Biomedical Instrumentation, Orient Longman Publishers (2000) (UNIT V)

Reference Books:

1. R.Anandanatarajan, "Biomedical Instrumentation", PHI Learning, 2009.
2. M. Arumugam, "Biomedical Instrumentation", Anuradha Agencies Publishers, Vidayal Karuppar, 612 606, Kumbakonam, R.M.S: 1992

Course Outcomes: At the end of this course, Students will be able to:

1. Analyze the basics concepts of Transducers and circuit based on transduction.
2. Illustrate the various types of reference electrodes.
3. Analyze the different types of ECG, EEG, EMG, EOG, ERG signals.
4. Illustrate the procedure of principle, types function and working of dialysis.
5. Analyze the Physiological effects of electrical currents and radiation hazards.

BEC-551	INTEGRATED CIRCUITS LAB	0L:0T:2P	1 Credit

SUGGESTIVE LIST OF EXPERIMENTS:

1. Design the following using Op-Amp: ***(Through Virtual Lab Link 1)***
   a) A unity gain amplifier.
   b) An inverting amplifier with a gain of "A".
   c) A non-inverting amplifier with a gain of "A"
2. Study and design Log and antilog amplifiers.
3. Voltage to current and current to voltage convertors.
4. Second order filters using operational amplifier for: ***(Through Virtual Lab Link 1)***
   a) Low pass filter of cutoff frequency 1 KHz.
   b) High pass filter of frequency 12 KHz.
5. Realization of Band pass filter with unit gain of pass band from 1 KHz to 12 KHz.
6. Study and design voltage comparator and zero crossing detectors.
7. Function generator using operational amplifier (sine, triangular & square wave).
8. Design and construct astable multivibrator using IC 555 and
   a) Plot the output waveform
   b) Measure the frequency of oscillation ***(Through Virtual Lab Link 2)***
9. Design and construct a monostable multivibrator using IC 555 and
   a) Plot the output waveform
   b) Measure the time delay ***(Through Virtual Lab Link 2)***
10. Implement Schmitt Trigger Circuit using IC 555. ***(Through Virtual Lab Link 2)***
11. Implement voltage-controlled oscillator using IC566 and plot the waveform. ***(Through Virtual Lab Link 2)***
12. Study and design ramp generator using IC 566.

Virtual Lab Link:

1. http://vlabs.iitkgp.ernet.in/be/exp17/index.html
2. http://hecoep.vlabs.ac.in/Experiment8/Theory.html?domain=ElectronicsandCommunications&lab=Hybrid%20Electronics%20Lab

Available on: http://www.vlab.co.in/broad-area-electronics-and-communications

Course Outcomes:

At the end of this course students will demonstrate the ability to:

1. Design different non-linear applications of operational amplifiers such as log,antilog amplifiers and voltage comparators.
2. Explain and design different linear applications of operational amplifiers such as filters.
3. Demonstrate the function of waveforms generator using op-Amp.
4. Construct multivibrator and oscillator circuits using IC555 and IC566 and perform measurements of frequency and time.
5. Design and practically demonstrate the applications based on IC555 and IC566.

BEC-552	MICROPROCESSOR & MICROCONTROLLER LAB	0L:0T:2P	1 Credit

SUGGESTIVE LIST OF EXPERIMENTS:

1. ***1.*** Write a program using 8085 Microprocessor for Decimal, Hexadecimal addition and subtraction of two Numbers. ***(Through Virtual Lab Link)***
2. Write a program using 8085 Microprocessor for addition and subtraction of two BCD numbers. ***(Through Virtual Lab Link)***
3. ***3.*** To perform multiplication and division of two 8 bit numbers using 8085. ***(Through Virtual Lab Link)***
4. To find the largest and smallest number in an array of data using 8085 instruction set.
5. To write a program using 8086 to arrange an array of data in ascending and descending order. ***(Through Virtual Lab Link)***
6. To convert given Hexadecimal number into its equivalent ASCII number and vice versa using 8086 instruction set.
7. To convert given Hexadecimal number into its equivalent BCD number and vice versa using 8086 instruction set.
8. To interface 8253 programmable interval timer and verify the operation of 8253 in six different modes.
9. To write a program to initiate 8251 and to check the transmission and reception of character.
10. Serial communication between two 8085 through RS-232 C port.
11. Write a program of Flashing LED connected to port 1 of the 8051 Micro Controller
12. Write a program to generate 10 kHz square wave using 8051.
13. Write a program to show the use of INT0 and INT1 of 8051.
14. Write a program for temperature & to display on intelligent LCD display.

Virtual Lab Link: http://vlabs.iitb.ac.in/vlabs-dev/labs_local/microprocessor/labs/explist.php

Available on: http://www.vlab.co.in/broad-area-electronics-and-communications

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Use techniques, skills, modern engineering tools, instrumentation and software/hardware appropriately to list and demonstrate arithmetic and logical operations on 8 bit data using microprocessor 8085.
2. Examine 8085 & 8086 microprocessor and its interfacing with peripheral devices.
3. State various conversion techniques using 8085 & 8086 and generate waveforms using 8085.
4. Implement programming concept of 8051 Microcontroller.
5. Design concepts to Interface peripheral devices with Microcontroller so as to design Microcontroller based projects.

BIC-551	INSTRUMENTATION & MEASUREMENT LAB	0L:0T:2P	1 Credit

SUGGESTIVE LIST OF EXPERIMENTS:

1. Study of semiconductor diode voltmeter and its use as DC average responding AC Voltmeter.
2. Study of L.C.R. Bridge and determination of the value of the given components.
3. Characteristics of Thermocouples and RTD.
4. Study of the following transducer (i) PT-100 Transducer (ii) J-Type Transducer (iii) K-Type Transducer (iv) Pressure Transducer
5. Measurement of phase difference and frequency using CRO (Lissajous Figure)
6. Characteristics of LDR, Photo Diode, and Phototransistor:
   (i) Variable Illumination.
   (ii) Linear Displacement
7. Characteristics of LVDT.
8. Study of the transistor tester and determination of the parameters of the given transistors
9. Design and Test a signal conditioning circuit for any transducer.
10. Implementation of Color Sensor for differentiating frequencies

Through Virtual Lab:

11. Measurement of low resistance Kelvin's double bridge.
12. To measure unknown capacitance of small capacitors by using Schering's bridge.
13. To measure unknown Inductance using Hay's bridge.
14. Measurement of capacitance by De Sauty Bridge.

Virtual Lab Link: http://vlabs.iitkgp.ernet.in/asnm/#

Available on: http://www.vlab.co.in/broad-area-electronics-and-communications

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Demonstrate voltage measurement using AC voltmeter and semiconductor diodevoltmeter.
2. Measure the unknown resistance, capacitance and inductance using LCR Bridge,Kelvin double bridge, Schering bridge, Hay's bridge, De sauty bridge.
3. Practically demonstrate the different types of transducers like J-type, K-type, PT-100and RTD.
4. Interpret frequency and phase difference from Lissajous figure.
5. Interpret hybrid parameters of transistor and demonstrate different transducer like LDRand LVDT.

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B.Tech 3rd Year
VI Semester
Syllabus

BIC-601	EMBEDDED SYSTEMS DESIGN	3L:1T:0P	4 Credits

Unit	Topics	Lectures
I	**Advanced concepts in 8051 architecture:** Review of 8051 architecture, concept of synchronous serial communication, SPIand I2C communication protocols, study of SPI port on 89LP 51RD2, study of SAR ADC/DAC MCP3304 / MCP 33, interfacing concepts for SPI based ADC/DAC, study of watchdog timer, study of PCA timer in different modes likecapture mode, PWM generation mode, High speed output toggle mode Embedded 'C' programming for the above peripherals.	8
II	**MSP430x5x Microcontroller:** series block diagram, address space, on-chip peripherals (analog and digital), and Register sets. Instruction set, instruction formats, and various addressing modes of 16-bit microcontroller; Sample embedded system on MSP430 microcontroller. Memory Mapped Peripherals, programming System registers, I/O pin multiplexing, pull up/down registers, GPIO control. Interrupts and interrupt programming.	8
III	**Introduction to Embedded Systems:** Describe what an embedded system is andits main components, Outline the different options available for buildingembedded systems, Explain the benefits, functions, and attributes of embedded systems, Examine the constraints specific to embedded systems and their impact **Introduction to the Arm Cortex-M4 Processor** Architecture, Identify key features of Arm architectures and processors, Explain the features and layout ofthe Arm Cortex-M4processor. Explain the structure and purposeof specific registers in the Arm Cortex-M4 processor. **Introduction to Arm Cortex-M4 Programming**, Compare the C and Assembly programming languages, Explain program-generation flow, including compilation and program images, Describe and compare different data formats and how they are stored in memory, Explain how mixed assembly and C programming can be performed, Introduction to the Mbed Platform and CMSIS.	8
IV	**Digital Input and Output (IO):** Explain the relationship between electricalvoltages and logic values, Describe the key features of GPIOs (General PurposeI/O pins) and how they can be used to control peripherals, Explain the keyelements of GPIO design in relation to microcontrollers. Compare register-level GPIO programming to GPIO programming with the Mbed API. Interrupts andLow Power Features: Interrupts and Low Power Features, Serial Communication **Introduction to the Internet of Things (IoT):** Describe the concepts of IoT and understand the key elements of an IoT device, Outline the evolution of IoT,Describe the main technologies that enable IoT, Identify the key challenges facing IoT systems, and Evaluate the opportunities and risks that emerge with IoTadoption.	8
V	**Hardware Platforms for IoT:** Identify the concepts of hardware platform and the factors influencing its design, Differentiate between various types of memory, Explain the principles of sensors and the role of I/O **IoT Connectivity:** Identify the concept of Bluetooth technology, Identify keyfeatures of the Bluetooth and Bluetooth Low Energy protocols, Explain how a Bluetooth connection is secured, Outline the new features that are introduced in the Bluetooth 5 specification, Explain the architecture and protocol stack used inZigBee.	8

Text Books:

1. Mazidi Ali Muhammad, Mazidi Gillispie Janice, and Mc Kinlay Rolin D "The 8051 Microcontroller and Embedded Systems using Assembly and C", Pearson Publication,2006
2. John H Davies, "MSP430 Microcontroller Basics" Newnes Publication,2008.
3. Embedded Systems Fundamentals on Arm Cortex-M based Microcontrollers: A Practical Approach by Alexander G. Dean https://www.arm.com/resources/education/textbooks/efficient- embedded-systems

Reference Books:

1. TI MSP430x5xx and MSP430x6xx Family User's Guide, Revised 2018.
2. The Definitive Guide to ARM® Cortex®-M3 and Cortex®-M4 Processors, Third Edition by Joseph Yiu.
3. Cortex-A Series Programmer's Guide for ARMv7-A by Arm from http://infocenter. arm.com/ help/topic/com.arm.doc.den0013d/index.html
4. White Paper: Cortex-M for Beginners - An overview of the Arm Cortex-M processor family and comparison: https://community.arm.com/developer/ip-products/processors/b/processors-ip-blog/posts/white-paper-cortex-m-for-beginners-an-overview-of-the-arm-cortex-m-processor-family- and-comparison.

Course Outcomes: At the end of this course, Students will be able to:

1. Explain the advance concept of 8051 architectures and AVR family architecture andcompare them for different applications.
2. To demonstrate the basics of MSP430x5x Microcontroller
3. To execute the I/O interfacing and peripheral devices associated with MicrocontrollerSoC (system on chip).
4. Explain the advance concept Arm Cortex-M4 Processor Architecture.
5. Demonstrate the ability to do Demonstrate the basics of Embedded Systems, IoT and itsapplication and design IoT based projects on Arm based development boards

BEC-602	CONTROL SYSTEM	3L:1T:0P	4 Credits

Unit	Topics	Lectures
I	**Introduction to Control Systems:** Basic Components of a control system, Feedback and its effect, types of feedback control systems. Block diagrams Reduction and signal flow graphs, Modeling of Physical systems: electrical networks, mechanical systems elements, free body diagram, analogous Systems, sensors and encoders in control systems, modeling of armature controlled and field controlled DC servomotor.	8
II	**State-Variable Analysis:** Introduction, vector matrix representation of state equation, state transition matrix, state-transition equation, relationship between state equations and high- order differential equations, relationship between state equations and transfer functions, Decomposition of transfer functions, Controllability and observability, Eigen Value and Eigen Vector, Diagonalization.	8
III	**Time domain Analysis of Control Systems:** Time response of continuous data systems, typical test signals for the time response of control systems, unit step response and time-domain specifications, time response of a first order system, transient response of a prototype second order system, Steady-State error, Static and dynamic error coefficients, error analysis for different types of systems.	8
IV	**Stability of Linear Control Systems:** Bounded-input bounded-output stability continuous data systems, zero-input and asymptotic stability of continuous data systems, Routh Hurwitz criterion, Root-Locus Technique: Introduction, Properties of the Root Loci, Design aspects of the Root Loci.	8
V	**Frequency Domain Analysis:** Resonant peak and Resonant frequency, Bandwidth ofthe prototype Second order system, effects of adding a zero to the forward path, effects of adding a pole to the forward path, polar plot, Nyquist stability criterion, stability analysis with the Bode plot, relative stability: gain margin and phase margin.	8

Text Book:

1. I. J. Nagrath & M. Gopal, "Control System Engineering", 6th Ed. New Age International Publishers, 2018
2. B.C. Kuo & Farid Golnaraghi, "Automatic Control Systems", 9th Edition, John Wiley India,2008

Reference Books:

1. (Schaums Outlines Series) Joseph J. Distefano III, Allen R. Stubberud, Ivan J. Williams, "Control Systems", 3rd Edition, TMH, Special Indian Edition, 2010.
2. A. Anand Kumar, "Control Systems", Second Edition, PHI Learning private limited, 2014.
3. William A. Wolovich, "Automatic Control Systems", Oxford University Press, 2011.

Course Outcomes: At the end of this course, Students will be able to:

1. Describe the basics of control systems along with different types of feedback and its effect. Additionally, they will also be able to explain the techniques such as block diagrams reduction,signal flow graph and modelling of various physical systems along with modellingof DC servomotor.
2. Explain the concept of state variables for the representation of LTI system.
3. Interpret the time domain response analysis for various types of inputs along with the timedomain specifications.
4. Distinguish the concepts of absolute and relative stability for continuous data systems alongwith different methods.
5. Interpret the concept of frequency domain response analysis and their specifications.

BIC-602	INDUSTRIAL MEASURING INSTRUMENTS	3L:1T:0P	4 Credits

Unit	Topics	Lectures
I	**Generalized configurations, functional descriptions and performance characteristics of measuring instruments:** General concepts and terminology ofmeasurement systems, transducer classification, general input-output configuration, static and dynamic characteristics of a measurement system, Statistical analysis of measurement data. Standards and Calibration. **Displacement measurement:** Resistive potentiometers, Digital displacementtransducers, Mechanical fly ball angular velocity sensor, Mechanical revolution counters and timers, stroboscopic method	6
II	**Force and Pressure Measurement:** Standards & calibration; basic methods of force measurement; Characteristics of elastic force transducer-Bonded strain gauge, differential transformer, Piezo electric transducer. Units of pressure; deadweight gauges & manometer and its types, Bellows and force balance type sensors, Bourden guage, Piezoelectric, Capacitive and Inductive Pressure pickups.	10
III	**Flow measurement:** Differential pressure flowmeters: Bernoulli's theorem: pitot tube, orifice, venturi, flow nozzle, Hot wire and hot film anemometers, variable area meters (rotameter), meters, Electromagnetic flowmeters, Ultrasonic flowmeters, Drag force flow meter, Vortex shedding flow meters. Measurement of level, Float type gauge, purge method, differential pressure method, conductive and capacitive method; electromechanical method.	8
IV	**Temperature measurements:** Standards and calibration, thermal expansion methods, bimetallic thermometer, thermocouple, reference junction considerations, special materials, configuration & techniques, Measurement of thermocouple output, Electrical resistance sensors and thermistors, Radiation thermometers.	8
V	**Miscellaneous Measurements: Viscosity, Density and Vacuum:** **Measurement of Viscosity:** Definitions, units, Newtonian and Newtonian behaviour, measurement of viscosity using laboratory viscometers, industrial viscometers. Viscometer selection and application. **Measurement of Density:** Definitions, units, liquid density measurement, gas densitometers, its application and selection, **Measurement of Vacuum:** Mcleod gauge, Pirani gauge, Knudsen gauge and Ionization gauge	8

Text Books:

1. E. O. Doebelin, "Measurements systems: Applications and Design", 4th Edition, Tata McGrawHill.
2. B. C. Nakra and K. K. Chaudhry, "Instrumentation: Measurements & Analysis" Tata McGrawHill
3. J.G. Joshi, Electronics Measurement and Instrumentation, Khanna Publishing House, Delhi.

Reference Books:

1. A.K. Sawhney, "A Course in Electrical and Electronic Measurements and Instrumentation"Dhanpat Rai Publications., 19th Edition.
2. Bela G. Liptak, "Process Measurement and Analysis, Vol. 1", CRC Press

Course Outcomes: At the end of this course, Students will be able to:

1. Describe the basic fundamentals, terms, and characteristics of measurement system.
2. Explain the working principle of various transducers used for the measurement of force and pressure.
3. Recognize the physics of pressure, temperature, level and flow measurement used to control dynamics of processes.
4. Assemble commonly used temperature measurement devices through proper selection, identification, design, installation and principle of operation in industries.
5. Develop critical and creative thinking to bring the technology, problem-solving skills in trouble shooting problems with the measurement and control of industrial instrumentationwork.

BIC-061	DIGITAL MEASUREMENT TECHNIQUES	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Philosophy of digital measurements**: Digital Time Measurement Techniques: Measurement of time interval between two events, Error in time interval measurement, Vernier technique for small time measurement, Measurement of time interval with constraints, Measurement of periodic time, phase, Time interval between two events defined by voltage levels, Capacitance, Quality factor of ringing circuit, Decibel meter	8
II	**Digital Frequency Measurement Techniques:** Measurement of frequency, Ratio of two frequencies, Product of two frequencies, High frequency, average Frequency difference, Deviation of power frequency, Peak frequency. Fast low-frequency measurement.	8
III	**Digitally Programmable Circuits:** Single mode switching, Group mode switching, Resistors, Potentiometers, Amplifiers, Schmitt trigger, Dual polarity gain amplifiers. Programmable gain amplifier with dual output, Two stage programming, Programmable Biquads.	8
IV	**Digital to Analog Converters:** Output Input relation, DACs derived from programmable gain amplifiers, Weighted-resistor DAC, Weighted current DAC, Weighted reference voltage DAC, Ladder DAC, Switches.	8
V	**Digital Voltage Measurement Techniques:** Sampling theorem, Time-division multiplexing, Quantization, Indirect type A/D converters, Direct type A/D converters, Input circuitry of a digital voltmeter.	8

Text Books:

1. "Digital Measurement Techniques", T. S. Rathore, by Narosa Publishing House, 1996

Reference Books:

1. A.K. Maini, All in One Electronics Simplified, Khanna Publishing House, Delhi

Course Outcomes: At the end of this course, Students will be able to:

1. To describe the basic knowledge of Digital Measurement Technique.
2. To demonstrate the concepts of DMT involved in term of frequency.
3. To explain the concepts of programmable circuits.
4. To analyze the performance of digital to analog converters.
5. To apply the concept of voltage measurement in DMT.

BEC-062	DATA COMMUNICATION NETWORKS	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Introduction to Networks & Data Communications:** Goals and Applications of Networks, The Internet, Protocols & Standards, Layered Tasks, OSI reference Model, TCP / IP, Addressing, Line Coding Review.	8
II	**Physical Layer**: Transmission Media- Guided and unguided, Network Topology Design, **Data Link Layer**: Error detection and Correction, Framing, Flow and Error Control Protocols, Noiseless Channel and Noisy Channel Protocol, HDLC, Point-to-Point Protocol	8
III	**Multiple Access:** RANDOH, CDMA, CSMA/CD, CSMA/CA, Controlled Access, Channelization Wired LANs: IEEE Standards, Standard Ethernet, Fast Ethernet, Gigabit Ethernet, Wireless LAN IEEE 802.11, Bluetooth IEEE 802.16.	8
IV	**Network Layer:** Design Issues. Routing Algorithms. Congestion control Algorithms. Internetworking –TCP/IP, IP Packet, IPv4 and IPv6 Protocols, IPV4 Addresses, Connecting Devices, Virtual LAN IPV6 Addresses.	8
V	**Transport Layer Protocol:** UDP and TCP, ATM, Cryptography, Network Security**,** Session Layer-Design issues. **Application Layer:** File Transfer, Electronic mail, HTTP, WWW, SMTP, Cryptography, Network Security.	8

Text Books:

1. B. A. Forouzan, "Data Communications and Networking", 5th Edition, TMH, 2017.

Reference Books:

1. S. Tanenbaum, "Computer Networks", 4th Edition, Pearson, 2013.
2. W. Stallings, "Data and Computer Communication", 8th Edition, Pearson, 2007.

Course Outcomes: At the end of this course, Students will be able to:

1. Identify the issues and challenges in the architecture of a network.
2. Analyze the services and features of various protocol layers in data layer.
3. Demonstrate the knowledge of multiple access to design a access technique for a particular application.
4. Realize protocols at different layers of a network hierarchy.
5. Recognize security issues in a network and various application of application layer.

BIC-062	BIO-MEDICAL SIGNAL PROCESSING	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Introduction to Bio-Medical Signals:** Classification, Acquisition and Difficulties during Acquisition. Basics of Electrocardiography, Electroencephalography, Electromyography & electro-retinography Role of Computers in the Analysis, Processing, Monitoring & Control, and image reconstruction in bio-medical field.	8
II	**ECG:** Measurement of Amplitude and Time Intervals, QRS Detection (Different Methods), ST Segment Analysis, Removal of Baseline Wander and Power line Interferences, Arrhythmia Analysis, Portable Arrhythmia Monitors.	8
III	**Data Reduction:** Turning Point algorithm, AZTEC Algorithm, Fan Algorithm, Huffman and Modified Huffman Coding, Run Length Coding.	8
IV	**EEG:** Neurological Signal Processing, EEG characteristic, linear prediction theory, Sleep EEG, Dynamics of Sleep/Wake transition. Study of pattern of brain waves, Epilepsy-Transition, detection and Estimation. EEG Analysis By Spectral Estimation: The Bt Method, Periodogram, Maximum Entropy Method & AR Method, Moving Average Method. The ARMA Methods, Maximum Likelihood Method.	8
V	**EP Estimation:** by Signal Averaging, Adaptive Filtering: - General Structures of Adaptive filters, LMS Adaptive Filter, Adaptive Noise Cancelling, Wavelet Detection: - Introduction, Detection by Structural features, Matched Filtering, Adaptive Wavelet Detection, Detection of Overlapping Wavelets.	8

Text Books:

1. Willis J. Tomkin, "Biomedical Digital Signal Processing", PHI.
2. D. C. Reddy, "Biomedical Signal Processing", McGraw Hill
3. Crommwell, Weibel and Pfeifer, "Biomedical Instrumentation and Measurement", PHI

Reference Books:

1. Arnon Cohen, "Biomedical Signal Processing (volume-I)", Licrc Press.
2. Rangaraj M. Rangayyan, "Biomedical Signal Analysis A Case Study Approach", John Wiley and Sons Inc.
3. John G. Webster, "Medical instrumentation Application and Design", John Wiley & Sons Inc.

Course Outcomes: At the end of this course, Students will be able to:

1. To understand the basics of biomedical signals.
2. Analyses of ECG signals.
3. To Analyze the data reduction techniques.
4. Analyses of EEG signals.
5. To understand the advance techniques of signal processing

BIC-063	MICROCONTROLLERS AND SENSORS FOR IOT DEVICES	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Fundamentals of IOT:** Introduction to IoT, IoT Design Methodology, Design Principles for Connected Devices: Calm and Ambient Technology, Internet Communications overview, IoT enabled Technologies – Wireless Sensor Networks, Cloud Computing.	8
II	**Data Acquisition with IOT Devices:** Prototyping embedded devices: Sensors and Actuators, Electronics and Power Components, Working mechanism of range of sensors and data acquisition, Data processing and analysis, Wireless Data transmission, IoT Point to point communication technology, IoT Communication Patterns, Selection of Wireless technologies (6LoWPAN, Zigbee, WIFI, BLE,NFC, LORA, Li-Fi, Widi).	8
III	**MSP430 Microcontrollers:** Functional Block diagram, Addressing modes, Instruction Set, Examples, Reflections on the CPU and Instruction Set, DPR for M.Tech IoT and Sensor Systems 2022 19 Exceptions, Clock system. Development: Development Environment, Access to the Microcontroller for Programming and Debugging, Demonstration Boards, Examples. Functions, Interrupts, and Low-Power Modes. Functions and Subroutines, Interrupts, ISR, Low-Power Modes of Operation.	8
IV	**Sensor Interfacing and Applications:** Sensors interfacing: Sensors interfacing techniques-Port Programming, ADC, SPI thermometer, I2C thermometer, PWM generation and demodulation, DTH11Cloud interfacing: Interfacing and data logging withcloud: Thing speak. **IoT's Applications** – Smart Farming: Smart Greenhouse, Drones for pesticides, Smart Energy Meters, Home automation, Smart Grid and Solar Energy Harvesting, Intelligent Parking, IoT for Healthcare, and weather analysis.	8
V	**Security and Privacy in IoT:** The problem- "Identity of Things", Device On-Boarding, Symmetric Encryption Standards: Symmetric Encryption (DES, AES 128), Hashing, Authentication, CCMP Authentication and Encryption protocol, Non-Symmetric Encryption Standards: Diffie Hellman (principle, Man in the Middle attack), RSA, Digital Signature and Authentication.	**8**

Text Books:

1. A Beginner's Guide to Internet of Things Security: Attacks, Applications, B.B. Gupta, Aakansha Tiwari, CRC Press, Taylor and Francis Group, 2020.
2. Donald Norris, "The Internet of Things: Do-It-Yourself at Home Projects for Arduino, Raspberry Pi and BeagleBone Black", McGraw Hill, 2015.
3. John H. Davies, "MSP430 Microcontroller Basics", 2011, 2nd ed., Newnes publishing, New York.

Reference Books:

1. Peter Waher, 'Learning Internet of Things', Packt Publishing, 2015.
2. A First Course in Fuzzy and Neural Control by Nguyen, Prasad, Walker, and Walker. CRC 2003.
3. Jacob Fraden, "Hand Book of Modern Sensors: physics, Designs and Applications", 2014, 4th ed., Springer, New York.

Course Outcomes: At the end of this course, Students will be able to:

1. Introduction to IoTs enabled technologies.
2. Analyze the mechanism of Data Acquisition and processing.
3. Understand data classification, process of big data technology, user roles and Detects myths in big data
4. Analyze software and management tools for Industrial applications of IoTs
5. Understand the Real-World design constraints and challenges.

BIC-651	EMBEDDED SYSTEM DESIGN LAB	0L:0T:2P	1 Credit

SUGGESTIVE LIST OF EXPERIMENTS:

Part A: Based on ARM Process:

1. To develop and verify the interfacing ADC and DAC with LPC 2148 Arm Micro Controller.
2. Interfacing of LED and PWM with Micro Controller. (ARM-) using embedded C program.
3. Interfacing of serial port with Am processor using embedded C-program.
4. Interfacing of key board and LCD with Arm processor using embedded C-Program.
5. To develop and verify Embedded C program mailbox using ARM.
6. To implement zigbee protocol with ARM program.
7. Implement the lighting and winking LEDs of the ARM I/O port via programming.
8. ARM programming in C language using KEIL IDE.
9. Demonstrate the TIMING concept of real time application using RTOS on ARM microcontrollerkit.
10. Demonstrate the Multi-Tasking concept of real time application using RTOs on ARMmicrocontroller.
11. Demonstrate the RS 232 serial communication using RTOS on ARM microcontroller kit.
12. ISR (Interrupt Service Routine) programming in ARM based system with I/O port.

Part B: Based on MSP 430

1. Write a program for temperature & to display on intelligent LCD display.
2. Write a program to generate a Ram waveform using DAC with micro controller.
3. Write a program to Interface GPIO port in C using MSP430 (blinking LEDs, push buttons)
4. Write a program Interface potentiometer with GPIO.
5. Write a program of PWM based Speed Controller of Motor controlled by potentiometerconnected to GPIO.
6. Write a program of PWM generation using Timer on MSP430 GPIO.
7. Write a program to Interface an accelerometer.
8. Write a program using USB (Sending data back and forth across a bulk transfer-mode USB connection.)

Part C: Virtual Lab Platform

https://www.soe.uoguelph.ca/webfiles/engg4420/EmbeddedSystemsAndLabsForARM-V1.1.pdf
https://profile.iiita.ac.in/bibhas.ghoshal/IEMB_2018/Lectures/ES_basics.pdf
https://nptel.ac.in/courses/108/102/108102045/

Practical Outcome The Student able to:

1. To understand the building blocks of embedded system.
2. To learn the concept of interfacing with different devices.
3. To relate the concept of memory map and memory interface.
4. To discover the characteristics of real time system and to validate the process using know input-output parameters.
5. To understand the basis work of microcontroller and learn the working.
6. Demonstrate knowledge of programs environment and executing variety of programs.

BEC-652	CONTROL SYSTEM LAB	0L:0T:2P	1 Credit

SUGGESTIVE LIST OF EXPERIMENTS:

1. Introduction to MATLAB Control System Toolbox.
2. Determine transpose, inverse values of given matrix.
3. Plot the pole-zero configuration in s-plane for the given transfer function.
4. Determine the transfer function for given closed loop system in block diagramrepresentation.
5. Create the state space model of a linear continuous system.
6. Determine the State Space representations of the given transfer function.
7. Determine the time response of the given system subjected to any arbitrary input.
8. Plot unit step response of given transfer function and find delay time, rise time, peaktime, peak overshoot and settling time.
9. Determine the steady state errors of a given transfer function.
10. Plot root locus of given transfer function, locate closed loop poles for different values ofk.
11. Plot bode plot of given transfer function. Also determine gain and phase margins.
12. Plot Nyquist plot for given transfer function. Also determine the relative stability bymeasuring gain and phase margin.

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Classify different tools in MATLAB along with the basic matrix operations used in MATLAB.
2. Evaluate the poles and zeros on s-plane along with transfer function of a given system.
3. Construct state space model of a linear continuous system.
4. Evaluate the various specifications of time domain response of a given system.
5. Appraise the steady state error of a given transfer function.
6. Examine the relative stability of a given transfer function using various methods such as root locus, Bode plot and Nyquist plot.

BIC652	**INDUSTRIAL INSTRUMENTATION LAB**	**0L:0T:2P**	**1 Credit**

SUGGESTIVE LIST OF EXPERIMENTS:

1. Instrumentation Amplifier: Design for specific gain and verification of CMRR.
2. Realization of PCM signal using ADC and reconstruction using DAC using 4-bit/8bitsystems. Observe the Quantization noise in each case.
3. Study of low noise and low frequency amplifier for biomedical application.
4. Design of temperature transmitter using RTD.
5. Design of cold junction compensation circuit.
6. Design of Linearization circuit for thermistor.
7. Design of pressure transmitter.
8. Performance evaluation of pressure gauges using Dead weight tester.
9. Measurement of level using capacitance probe, differential pressure transmitter.
10. Measurement of flow using orifice, electromagnetic and positive displacementflowmeters.
11. Study of PID controllers in flow measurement.
12. Measurement of solar energy using sensor.
13. Experiment using PLC Trainer Kits
14. Simulate and analyze the frequency domain measurement of electrical signals using spectrumanalyzer.
15. Range finding and object detection using detection sensor.
16. Measurement using various sensors and analyzing the output using VirtualInstrumentation Lab-VIEW software.

Course Outcomes: At the end of this course students will demonstrate the ability:

1. To design instrumentation amplifier.
2. To execute PCM technique and demonstrate the concept.
3. To design equipments related to temperature measurement.
4. To analyze the performance of various pressure gauges.
5. To conceptualize the principle of flow meters using various flow measurement instruments.